Hints: Number theory – 6

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Write $(f * g)(n) = \sum_{ab=n} f(a)g(b)$. Then commutativity is obvious, and associativity holds because both sides equal $\sum_{abc=n} f(a)g(b)h(c)$.

Hint 2. Inverse: solve $(f * g)(n) = \varepsilon(n)$ recursively. Set $g(1) = 1/f(1)$, and for $n > 1$ put $g(n) = -\frac{1}{f(1)} \sum_{d|n,\, d<n} f(n/d)g(d)$. Multiplicativity: for coprime m, n, the divisors of mn are exactly the products $d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$.

Hint 3. Inverses: $1^{-1} = \mu$ (Theorem 2). Since $d = 1 * 1$, $d^{-1} = \mu * \mu$, which is multiplicative with $p \mapsto -2$, $p^2 \mapsto 1$, $p^k \mapsto 0$ for $k \geqslant 3$. Since $\sigma = \mathrm{id} * 1$ and $\mathrm{id}^{-1} = \mu \cdot \mathrm{id}$, $\sigma^{-1} = (\mu\,\mathrm{id}) * \mu$, with $p \mapsto -(p+1)$, $p^2 \mapsto p$, $p^k \mapsto 0$ for $k \geqslant 3$.

Problem 2

Hint 1. Grouping: the fractions $\frac{k}{n}$, $1 \leqslant k \leqslant n$, written in lowest terms, have denominators $d \mid n$. Each denominator d occurs with exactly $\varphi(d)$ numerators. Multiplicativity: both sides of $\sum_{d|n} \varphi(d) = n$ are multiplicative, and at p^k the left side telescopes.

Hint 2. For $\sum_{d|n} \mu(d) = [n = 1]$, use $(1-1)^{\omega(n)}$, or multiplicativity. Then $g = f * 1 \iff f = g * \mu$ by associativity. From $\varphi * 1 = \mathrm{id}$ you get $\varphi = \mu * \mathrm{id}$, i.e. $\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \prod_{p|n} (1 - \frac{1}{p})$.

Hint 3. The last two identities are $\mu * d = \mu * 1 * 1 = 1$ and $\mu * \sigma = \mu * 1 * \mathrm{id} = \mathrm{id}$.

Problem 3

Hint 1. Compute small cases. Primes work: $p(p+1) \equiv 1 \cdot 2 \pmod{p-1}$. Also $n = 1, 4, 6, 22$ work. The task is to show nothing else does.

Hint 2. If $p^2 \mid n$, then $p \mid \varphi(n)$ and $p \mid n\sigma(n)$, so $p \mid 2$. Hence $n = 2^a m$ with m odd and squarefree. If $a \geqslant 3$, then $4 \mid \varphi(n)$ and $4 \mid n$, which is impossible. So $a \leqslant 2$.

Hint 3. If m has two odd prime factors, then $4 \mid \varphi(n)$. But $\sigma(m) = \prod(p+1)$ is divisible by 4, so $n\sigma(n) \equiv 0 \not\equiv 2 \pmod{4}$.

Hint 4. The remaining cases are $n = 2, 4$, $n = 2p$ and $n = 4p$. For $n = 2p$, reducing mod $p-1$ gives $p - 1 \mid 10$. For $n = 4p$, it gives $p - 1 \mid 54$. Check the few candidates. Answer: $n = 1, 4, 6, 22$ and all primes.

Problem 4

Hint 1. $\sum_{n \leqslant x} d(n)$ counts lattice points (a, b) with $ab \leqslant x$. By the symmetry $a \leftrightarrow b$ it equals $2 \sum_{a \leqslant \sqrt{x}} \lfloor x/a \rfloor - \lfloor \sqrt{x} \rfloor^2$.

Hint 2. Insert $\lfloor x/a \rfloor = x/a + O(1)$ and $\sum_{a \leqslant y} \frac{1}{a} = \ln y + \gamma + O(1/y)$. All error terms are $O(\sqrt{x})$, and the main terms give $x \ln x + (2\gamma - 1)x$.

Hint 3. Coprime pairs: by Möbius inversion, the number of coprime pairs in $\{1, \ldots, N\}^2$ is $\sum_{d \leqslant N} \mu(d) \lfloor N/d \rfloor^2 = N^2 \sum_{d \leqslant N} \frac{\mu(d)}{d^2} + O(N \ln N)$.

Hint 4. The tail of $\sum \mu(d)/d^2$ is $O(1/N)$, so the probability tends to $\sum_{d\geqslant 1} \mu(d)d^{-2} = 1/\zeta(2)$. Only here does the value $\zeta(2) = \pi^2/6$ enter (Euler's evaluation, from Series); the rest uses only $\mu * 1 = \varepsilon$.

Problem 5

Hint 1. Euclid: if $2^k - 1$ is prime, then $\sigma(2^{k-1}(2^k - 1)) = (2^k - 1)2^k$ by multiplicativity.

Hint 2. Euler: write $n = 2^{k-1}m$ with m odd and $k \geqslant 2$. Then $(2^k - 1)\sigma(m) = 2^k m$, so $m = (2^k - 1)t$ and $\sigma(m) = 2^k t = m + t$. Since $t < m$ are both divisors of m, the divisors of m are only t and m. Hence $t = 1$ and m is prime.

Hint 3. Odd perfect n: for a prime power, $\frac{\sigma(p^k)}{p^k} < \frac{p}{p-1} \leqslant \frac{3}{2} < 2$. With two distinct odd primes, $\frac{\sigma(n)}{n} < \frac{p}{p-1} \cdot \frac{q}{q-1} \leqslant \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{8} < 2$.